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Numerical sensitive data recognition based on hybrid gene expression programming for active distribution networks

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ABSTRACT

Complex and flexible access mode, and frequent data interaction bring about large security risks to data transmission for active distribution networks. How to ensure data security is critical to the safe and stable operation of active distribution networks. Traditional methods, like access control, data encryption, and text filtering based on intelligent algorithms, are difficult to ensure the security of dynamically increased and high-dimensional numerical data transmission in active distribution networks. In this paper, we first propose a rough feature selection algorithm based on the average importance measurement (RFS-AIM) to simplify the complexity of data recognition. Then, we propose a sensitive data recognition function mining algorithm based on RFS-AIM and improved gene expression programming (SDR-IGEP) where population update operation is constructed by chromosome similarity based on the Jaccard coefficient. The operation avoids local convergence of the gene express programming by increasing individual diversity in the new population. Finally, we present a new incremental mining algorithm for a sensitive data recognition function based on global function fitting (ISDR-GFF) by using a grain granulation model for incremental datasets. The experimental results on IEEE benchmark datasets and real datasets show that the algorithms proposed in this paper outperform the state-of-the-art algorithms in terms of the average running time, precision, recall, F_1 index, accuracy, specificity and speedup on all experimental datasets.

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1. Introduction

As an important part of information security in active distribution networks, data security has received increasing attention from researchers in recent years [1,2]. However, data security in active distribution networks is more complicated than traditional distribution networks due to complex network architecture and frequent data interaction.

With the extensive application of advanced information and communication technologies such as wireless communication and Internet of Things in active distribution networks, active distribution network must face an increasingly serious threat of viruses, Trojan horses and hacker attacks from the Internet [3–6]. Especially with the continuous construction of a strong smart grid, the active distribution network has a more complex access environment, flexible and diverse access methods, a large number of intelligent access terminals and dynamically distributed massive access data. Notably, with the access of a large number of distributed loads in the active distribution network, the interaction between the power grid and the users is greatly enhanced, which makes it possible for the users' electricity data and monitoring data of various equipment state operations to be stolen, modified and injected with bad data during network communications. Therefore, how to ensure the confidentiality, integrity and non-repudiation of business data in active distribution network has become a research hotspot for data security protection in active distribution network.

Traditional data security protection methods, such as data encryption [7] and access control [8], have high requirements on the computing power, storage capacity, and network transmission bandwidth of the user and server in active distribution network. The most important thing is that the business data transmitted in the active distribution network cannot be identified at the content level. Therefore, these data security protection methods are passive protection. To better solve data security protection, many researchers proposed data intelligent filtering based on content recognition [9,10] which can better achieve content level security protection. Compared with passive protection measures,

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it belongs to proactive protection. The existing data content filtering algorithms, which are based on text classification, focus on text and cannot effectively solve the leakage of numerical data in a SCADA system, an AMI system or smart meters in active distribution networks.

Gene expression programming (GEP) which is first proposed by Candida [11] is a new evolution algorithm. GEP has powerful classification and function mining capabilities [12,13], which can solve the problem of sensitive data identification well. Therefore, in this paper, we propose a novel parallel numerical data recognition algorithm for active distribution network based on feature selection and improved gene expression programming, which combine the advantages of rough sets and gene expression programming, to better protect the security of numerical data transmission in active distribution networks. The major contributions of our work are listed as follows:

- To reduce the complexity of sensitive data recognition model, this paper proposes a rough feature selection algorithm based on average importance measurement (RFS-AIM). The purpose is to quantitatively analyze the importance of each feature after reduction to the final decision feature.
- On the basis of RFS-AIM, in this paper, we propose a sensitive data recognition function mining algorithm based on feature selection and improved gene expression programming (SDR-IGEP). By using the concept of chromosome similarity, the algorithm improves the genetic operation in the traditional GEP and prevents the GEP population from falling into a local optimum.
- Meanwhile, an incremental mining algorithm of sensitive data recognition function based on global function fitting (ISDR-GFF) is proposed to solve the increasing sensitive data recognition function mining in active distribution networks. This algorithm constructs the architecture of parallel function mining based on grain granulation. At the same time, the multi-population grafting operation based on population similarity is proposed to improve the population diversity on the computing nodes and to increase the convergence speed of the GEP population.
- Experimental results on IEEE benchmark datasets and real datasets show that the proposed algorithms in this paper outperformed the traditional other algorithms in terms of the precision, recall, F_1 index, accuracy and specificity of sensitive data recognition, the average running time and speedup.

The remainder of this paper is organized as follows. Section 2 introduces a detailed overview of the related work. Section 3 focuses on the rough feature selection algorithm based on average importance measurement. Section 4 proposes a sensitive data recognition function mining algorithm based on feature selection and improved gene expression programming. Section 5 designs an incremental mining algorithm of sensitive data recognition function based on global function fitting. To evaluate the performance of the proposed algorithm, experimental results and analyses on IEEE benchmark datasets and real load datasets are given in Section 6. Conclusions are remarked in the last section.

2. Related work

2.1. Data security of smart grid

The access of various distributed energy sources and flexible loads makes the interaction of various intelligent terminals and users in smart grid more and more frequent and the distribution of data leakage points more and more extensive. Due to the insecurity of smart grid, the data in the cyber and physical system of the distribution network must address numerous security attacks and threats. Attia et al. [14] presented an intrusion detection system architecture for detecting illegal attacks with a focus on data availability and integrity. Li et al. [15] gave two data driven attacks with noisy measurements and proposed a new data recovery algorithm. Yang et al. [16] designed a protector for each sensor in a wireless sensor network to decide whether to use the received data at each time. Anwar et al. [17] proposed an improved constriction factor particle swarm optimization algorithm based hybrid clustering technology to divide the nodes. Tan et al. [18] revealed the intrinsic relationships between data integrity attacks and real time electrical market operations, and gave a systematic online attack construction strategy. Teixeira et al. [19] analyzed data integrity attacks in smart distribution networks.

The security risks of smart grid will make the physical protection of business data more difficult, and the range of network attackers that target sensitive data will also continue to expand. Li et al. [1] proposed a data security transmission scheme by using quantum encryption for smart grid. Guan et al. [7] presented a new secure data acquisition scheme by using text encryption in smart grid. Tan et al. [20] analyzed the security threats of data generation, acquisition, storage and processing, and pointed out that encryption is an important means of protecting data security. Romdhane et al. [21] used homomorphic encryption scheme to protect customer data privacy in smart grid. Guan et al. [22] proposed a flexible data access control scheme based upon encryption for smart grid with renewable energy resources. Although encryption is a more effective method for data security protection, complex encryption algorithms also need to design complex key distribution and storage mechanisms, which will inevitably increase the time and space complexity of the encryption algorithm. Moreover, for big data in smart grid [2], frequent data interaction cannot ensure data security through complex encryption and decryption.

These researchers analyzed all kinds of network attacks for data such as data integrity attacks, injection attacks, and so on, in the smart grid or distribution network, but did not consider the situation of sensitive data leakage caused by data attacks. However, all types of sensitive data in an active distribution network, such as SCADA data, measurement data, and various equipment failure data, are the core foundations of an active distribution network operation. Once data are stolen, the subsequent result is likely to threaten the safe and stable operation of active distribution networks.

2.2. Data recognition and filtering

Data filtering is essentially a classification based on sensitive data recognition. Gu et al. [10] proposed a multisource cross-domain classification algorithm based on the logistic regression and the existing consensus measure to solve cross-domain classification for the target domain. Wang et al. [23] presented a novel classification algorithm for network data to build a generative model for the network data. Experimental results showed that the proposed algorithm had high classification precision. Haq et al. [24] analyzed the special properties of hyperspectral data and proposed a minimization based sparse representation classification method to solve the problem of the high cost of the analysis. Extensive experiments showed that the proposed algorithm had advantages over the state-of-the-art methods. Nabila et al. [25] gave a novel classification algorithm based on the particle swarm optimization algorithm, which used a new particle position update mechanism in order to solve mixed-attribute

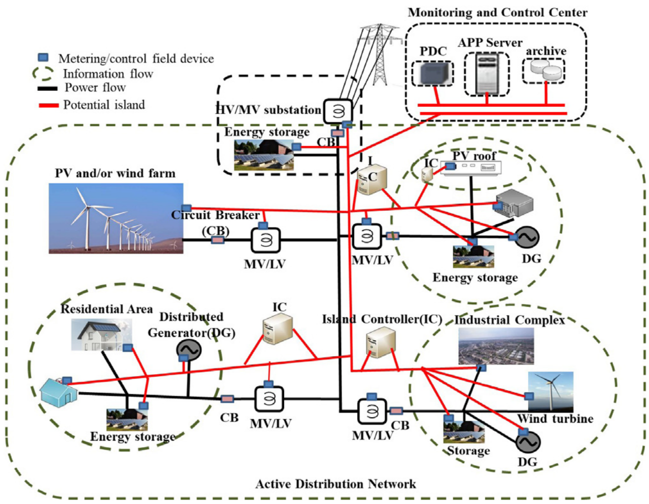

Fig. 1. Information flow and power flow in an active distribution network.

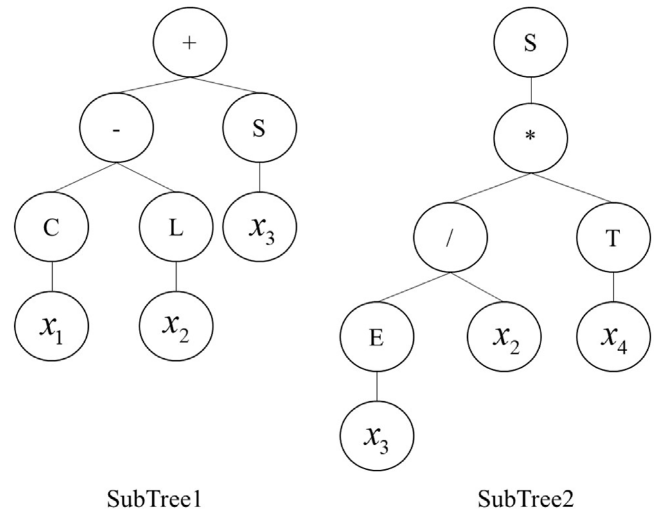

Fig. 2. The corresponding encoding scheme.

data. Lin et al. [26] proposed a new similarity measurement in text classification to better compute the similarity between two documents. Experimental results showed that the performance of the proposed algorithm was better than those of the existing methods. Li et al. [27] proposed a new text classification framework that used fuzzy formal concept analysis to avoid the adverse effect of ambiguous terms. Lee et al. [28] gave a fuzzy algorithm for multilabel text classification to overcome the curse of the dimensionality problem. Liu et al. [29] proposed a classification algorithm of defect texts based on a convolutional neural network for power equipment. Experimental results showed that the proposed algorithm could effectively reduce the error rate of text classification. Li et al. [30] proposed a novel model for relevance feature discovery in text mining to effectively use large scale patterns. Experimental results showed that the proposed algorithm outperformed the existing methods. Liu et al. [31] designed a semisupervised learning method with universum learning based on boosting. Experimental results indicated that the proposed algorithm outperformed the existing methods for universum examples.

The methods of data classification proposed in the above literatures have solved many problems in text classification well and achieved good results. However, these algorithms use text as the research object, and in addition to text in the active distribution network, there is a large amount of state monitoring, fault information, measurement and other numerical data related to the operating state of the active distribution network. And many of these data are numerical data, it is difficult to effectively identify these numerical data based on traditional text classification methods. At the same time, with the continuous enrichment of data monitoring methods in active distribution networks, data collection becomes more and more frequent, which results in a surge in data volume. Using the existing centralized data classification methods, it is also difficult to meet the real time and effective requirements of data processing.

3. Rough feature selection algorithm based on average importance measurement

An active distribution network is a classical cyber physical system. The source and interaction between the information flow and power flow in an active distribution network are shown in Fig. 1 [32]. Fig. 1 shows that the data sources in the active distribution network come from a wide range, mainly from power distribution SCADA systems or DMS systems, advanced metering facilities AMI, smart meters and so on. There are many types of data, including the status data of distribution lines, alarm data, power load data, log data, and information support system operation data. Obviously, for a complex active distribution network system, the larger the amount of detail in the system parameters and description, the more profoundly the system is understood. However, if many parameters are used to identify the system, it will take up a large amount of storage space and machine processing time. In power system control, the task is to use some parameters to control the system's operation. These parameters are also different in importance to the system identification or control. Furthermore, these parameters also contain a large number of redundant parameters. Therefore, to improve the efficiency and accuracy of the data analysis and processing, this paper proposes a rough feature selection algorithm that is based on the average importance measurement (RFS-AIM).

3.1. Definition

Let $K = (U, R)$ be a dataset. For each subset $X \subset U$, with the equivalence relation $R \subset ind(K)$, we have the following definitions and properties.

Definition 1. Let $K = (U, C \cup D)$ be a dataset, where C={c$_i$, $1 \leq i \leq n\}$ and D=$\{d_j, 1 \leq j \leq m\}$ represent the feature attributes and classification attributes of dataset K, respectively. Then $\delta_{c_i}(d_j) = \frac{|U - BN_{c_i}(d_j)|}{|U|}$ is called the importance measure of feature attribute c_i relative to classification attribute d_j, where $BN_{c_i}(d_j)$ is called the c_i boundary of d_j.

Definition 2. Let $K = (U, C \cup D)$ be a dataset, where C={c$_i$, $1 \leq i \leq n\}$ and D=$\{d_j, 1 \leq j \leq m\}$ represent the feature attributes and classification attributes of dataset K, respectively; $\delta_{c_i}(d_j) = \frac{|U - BN_{c_i}(d_j)|}{|U|}$ represents the importance measure of feature attribute c_i relative to classification attribute d_j. Then, $\overline{\delta_{c_i}(D)} = \frac{\sum_{j=1}^{m} \delta_{c_i}(d_j)}{|D|}$ is called the average importance measurement of feature attribute c_i relative to the classification attribute set D.

Property 1. *Suppose that $\overline{\delta_{c_p}(D)}$ and $\overline{\delta_{c_q}(D)}$ are the average importance measurement of feature attributes c_p and c_q relative to the classification attribute set D, respectively. If $\overline{\delta_{c_p}(D)} > \overline{\delta_{c_q}(D)}$, then the importance of feature attribute c_p is higher than c_q for the classification attribute set D.*

3.2. RFS-AIM

According to the above definitions, we propose a rough feature selection algorithm based on the average importance measurement (RFS-AIM) to reduce the abundant features in set. For the dataset $K = (U, C \cup D)$, where the feature attributes are represented as C={c_i, $1 \leq i \leq n$} and the classification attributes are represented as D={d_j, $1 \leq j \leq m$}, by calculating and sorting the importance measure of the feature attribute to the classification attribute, we can know an important degree order of all feature attributes to the classification attribute, Then, for the feature attribute, a certain number of feature attributes are selected according to the importance measure from large to small to form a new feature attribute set. Algorithm 1 is shown as follows.

Algorithm 1: RFS-AIM

Input: $K = (U, C \cup D)$,the number of features α ;
Output: Dataset after feature selection K';
1. $C' \leftarrow \emptyset$;
2. for i $\leftarrow$ 1 to n do
3. $Y_i \leftarrow U/c_i$;
4. end for
5. for j $\leftarrow$ 1 to m do
6. $X_j \leftarrow U/d_j$;
7. end for
8. for j $\leftarrow$ 1 to m do
9. for i $\leftarrow$ 1 to n do
10. $c_{i-}(X_j) \leftarrow \cup\{Y_i : Y_i \subset X_i\}$;
11. $c_i^-(X_j) \leftarrow \cup\{Y_i : Y_i \cap X_i \neq \emptyset\}$;
12. $BN_{c_i}(X_j) \leftarrow c_i^-(X_j) - c_{i-}(X_j)$;
13. $\delta_{c_i}(d_j) \leftarrow \frac{|U - BN_{c_i}(d_j)|}{|U|}$;
14. end for
15. end for
16. for i $\leftarrow$ 1 to n do
17. $\overline{\delta_{c_i}(D)} \leftarrow \frac{\sum_{j=1}^{m} \delta_{c_i}(d_j)}{|D|}$;
18. end for
/* Quickly sort all $\overline{\delta_{c_i}(D)}$, $i \in [1, n]$ in descending order */
19. $\overline{\delta_C(D)} \leftarrow QuickSort(\overline{\delta_C(D)}, left, right)$;
/* From the sorted $\overline{\delta_C(D)}$, take attributes to form a new feature set */
20. for k $\leftarrow$ 1 to α do
21. $C' \leftarrow C'.add(c_k)$;
22. end for
23. Return K'=(U,$C' \cup D$);

According Definition 2 and Property 1, the selection of feature attributes using the average importance measure of feature attributes is more efficient than the traditional feature reduction algorithms based on rough sets. The running time of Algorithm 1 focuses mainly on solving the importance measure of the feature attribute to the classification attribute and the quick sorting of the average importance measure. The average time complexity of solving the importance measure of the feature attribute to the classification attribute is approximately $O(mn)$, and the average time complexity of quickly sorting the average importance measure of all of the feature attributes to the classification attribute is approximately $O(n \log n)$.

4. Sensitive data recognition function mining algorithm based on feature selection and improved gene expression programming

Data from distribution SCADA systems, various status monitoring systems, AMIs, smart meters, and operational logs are the core assets of active distribution networks. These data come from various aspects such as power monitoring, equipment operation, and power consumption. In general, they are characterized by having a wide range of sources, having a large scale, and being composed of complex types. However, with the widespread use of various types of wireless communication technologies in active distribution networks, the vulnerabilities in various protocols and device operating systems will provide some possible opportunities for attackers to steal data. To prevent core data in the active distribution network from being stolen, traditional methods perform encryption processing at the data gateway, and then, they transmit the encrypted data to the primary station server by wire or wireless communication. Because various business systems generate enormous amounts of data in an active distribution network and the data processing and storage capacity of the data gateway are limited, it is impossible to encrypt all of the data. Therefore, based on the RFS-AIM, this paper proposes a sensitive data recognition function mining algorithm based on feature selection and improved gene expression programming (SDR-IGEP).

4.1. Improved gene expression programming

Gene expression programming (GEP) [11] is a typical evolution algorithm and mainly applied to function mining, classification and clustering etc. We know that traditional gene expression programming can easily fall into a local optimum [11,33,34]. Therefore, this paper proposes the concept of chromosome similarity based on the idea that the probability that a superior individual is higher when the father with the larger chromosome difference in the evolution of the biological genetics evolves to the next generation. By selecting two individuals with similar chromosome similarity from the father to perform transposition and recombination genetic operations, it is more likely that the next generation's individual fitness value will approach the optimal fitness value.

Definition 3. Let $X = \{x_1, x_2, \ldots, x_n\}$ and $Y = \{y_1, y_2, \ldots, y_n\}$. Then $J(X, Y) = \frac{|X \cap Y|}{|X \cup Y|}$ is called the Jaccard coefficient between X and Y, where $|X|$ represents the scale of set X. Furthermore, the chromosome similarity is proposed according to Definition 3.

Definition 4. Suppose that $G_1 = \{(h_1, t_1)|h_1 \in GHead, t_1 \in GTail\}$ and $G_2 = \{(h_2, t_2)|h_2 \in GHead, t_2 \in GTail\}$ are two chromosome sequences of equal length. Then $J(G_1, G_2) = \frac{|G_1 \cap G_2|}{|G_1 \cup G_2|}$ is called the chromosome similarity of two chromosome sequences.

From Definition 3, we know that the smaller the chromosome similarity of the two chromosome sequences, the greater the difference between the two chromosome sequences. According to Definitions 3 and 4, we obtain the following properties.

Property 2. *(1) $J(G_1, G_2) \in [0, 1]$. (2) when the chromosome sequences G_1 and G_2 are $\emptyset$, $J(G_1, G_2) = 0$. (3) When the chromosome sequences G_1 and G_2 are identical and not the empty set $\emptyset$, $J(G_1, G_2) = 1$.*

Proof. (1) Obviously, according to Definition 4, when the chromosome sequences G_1 and G_2 are $\emptyset$, $J(G_1, G_2) = 0$.

(2) When the chromosome sequences G_1 and G_2 are identical and not the empty set $\emptyset$, it can be seen from the definition of the set intersections and union calculation that $G_1 \cap G_2| = |G_1| = |G_2| = |G_1 \cup G_2|$ holds. Then, $J(G_1, G_2) = 1$ can hold according to Definitions 3 and 4.

(3) When the chromosome sequences G_1 and G_2 are not identical, it can be seen from the definition of the set intersections and union calculation that $G_1 \cap G_2| < |G_1 \cup G_2|$ holds.

In summary, $0 \leq J(G_1, G_2) \leq 1$. The Proof is thus completed.

Combined with the calculation of the chromosome similarity, when the gene expression programming performs the recombination and transposition genetic operation, the chromosome similarity calculation is first performed on the two randomly selected chromosomes. If the similarity of the two chromosomes is less than the threshold β, then the next step will be performed. Otherwise, the algorithm will continue to randomly select two chromosomes for the chromosome similarity calculation until the condition is met.

The description of the population update based on chromosome similarity (PUP-CS) is shown in Procedure 1.

Procedure 1: PUP-CS
Input: *Pop*, *popSize*, *p*, β;
Output: *newPop*;
1. newPop.size $\leftarrow 0$;
2. while newPop.size $\neq$ *popSize* do
/*Randomly select two chromosomes from the current population according to the probability *p**/
3. $\mathrm{G}_1 \leftarrow Pop.select(\mathrm{p}, \mathrm{Pop})$;
4. $\mathrm{G}_2 \leftarrow Pop.select(\mathrm{p}, \mathrm{Pop})$;
5. $J(G_1, G_2) \leftarrow \frac{|G_1 \cap G_2|}{|G_1 \cup G_2|}$;
6. if $J(G_1, G_2) < \beta$ then;
7. $newPop \leftarrow Pop.insert(G_1)$;
8. $newPop \leftarrow Pop.insert(G_2)$;
9. newPop.size$+=$2;
10. end if
11. end while
12. Return *newPop*;

Through the chromosome similarity calculation, the two chromosomes with the largest difference can be selected in the current population for the recombination and transposition genetic operation. This will greatly increase the diversity of the individuals in the next generation of the population, which could greatly improve the convergence speed and efficiency of the gene expression programming.

4.2. SDR-IGEP

Sensitive data recognition is essentially understood in terms of binary data classification. This approach finds a function model between the data features and the classification attribution for a set of labeled training data, and then, it performs automatic classification judgment on the data without sensing the labels, based on the function model. On the basis of RFS-AIM, this paper proposes a sensitive data recognition function mining algorithm based on feature selection and improved gene expression programming (SDR-IGEP). In the SDR-IGEP, the first thing to do is to encoding. The randomly generated GEP individuals are $+-SCLx_1x_2x_3x_5x_4x_3S*/TEx_2x_3x_4x_2x_5x_1$, and the corresponding encoding scheme in the SDR-IGEP is shown in Fig. 2.

To better perform SDR-IGEP, we first determine how to define the sensitive labels on the dataset. In this paper, we use the Tanimoto coefficient to define the sensitive data.

Definition 5. Let $X = \{x_1, x_2, \ldots, x_n\}$ and $Y = \{y_1, y_2, \ldots, y_n\}$ be two real vectors, where $x_i, y_i > 0$. Then, $T(X, Y) = \frac{X \bullet Y}{\|X\|^2 + \|Y\|^2 - X \bullet Y}$ is called the Tanimoto Coefficient between X and Y.

Definition 6. Suppose that $K^H = \{k_1^H, k_2^H, \ldots, k_n^H\}$ and $K^P = \{k_1^P, k_2^P, \ldots, k_n^P\}$ are sensitive and nonsensitive data, respectively, and $K = \{k_1, k_2, \ldots, k_n\}$ represents the data to be labeled, where $k_i, i \in [1, n]$ is the ith feature of K. If $T(K, K^{\mathrm{H}}) > T(K, K^P)$, then K is called sensitive data, and otherwise, K will be called nonsensitive data.

According to Definition 6, all of the collected data in the active distribution network are marked with sensitive labels, and then, the sensitive data recognition function mining is performed based on feature selection and improved gene expression programming. A description of Algorithm 2 is given as follows.

Algorithm 2: SDR-IGEP
Input: $K = (U, C \cup D)$, *popSize*, *maxFitness*, $\mathrm{P_m}$, $\mathrm{P_t}$, $\mathrm{P_r}$, β, p, α;
Output: *SenDRFunction*, *Pop*;
1. *Input*(K);
2. $\mathrm{K}' \leftarrow RFS - AIM(K, \alpha)$;
3. $Pop \leftarrow Init(popSize, \mathrm{K}')$;
4. EvaluatePop(Pop);
5. for i $\leftarrow$ 1 to *maxFitness* do
6. Pop $\leftarrow$ TournamentSelection(Pop,4);
7. Pop $\leftarrow$ Mutation(Pop,$\mathrm{P_m}$);
8. newPop $\leftarrow$ PUP $-$ CS(Pop, popSize, p);
9. Pop $\leftarrow$ ISTransposition($\mathrm{P_t}$, newPop);
10. Pop $\leftarrow$ RISTransposition($\mathrm{P_t}$, newPop);
11. Pop $\leftarrow$ GeneTransposition($\mathrm{P_t}$, newPop);
12. Pop $\leftarrow$ OneRecombination($\mathrm{P_r}$, newPop);
13. Pop $\leftarrow$ TwoRecombination($\mathrm{P_r}$, newPop);
14. Pop $\leftarrow$ GeneRecombination($\mathrm{P_r}$, newPop);
15. EvaluatePop(Pop);
16. end for
17. Return *SenDRFunction*, *Pop*

Through Algorithm 2, we obtain the corresponding sensitive data identification function $Sf = (t_1, t_2, \ldots, t_n|S)$, and the data to be identified is brought into $Sf = (t_1, t_2, \ldots, t_n|S)$ to calculate the corresponding classification attribute value; finally, the difference between the value and the real value is compared to determine whether the data to be identified belongs to the sensitive data to achieve sensitive data identification.

5. Incremental mining algorithm of the sensitive data recognition function based on global function fitting

The operation process of the active distribution network is not invariable. Because of the unstable output of the distributed generation and the strong load fluctuation, all types of electrical information and topological data in the operation process of the active distribution network change dynamically. Meanwhile, the active distribution network has a wide range of data sources and a wide geographical distribution. There are many system parameters that affect the safe and stable operation of the active distribution network, which makes the data in the active distribution network present the characteristics of having a large amount, high dimensionality and broad distribution.

Algorithm 2 requires geographically distributed data in the active distribution network to be stored in a centralized manner, and it does not have the capability of distributed processing. In the active distribution network, a large amount of geographically distributed data is uniformly transmitted to a server for processing, which increases the risk that the data be easily leaked during transmission and increases the network transmission band-width and the pressure of the analysis, processing and storage on the master server. Moreover, the operational data of the active distribution network also increases dynamically, and thus, the static sensitive data function model is difficult to adapt to the sensitive identification of the dynamically increased data.

Therefore, on the basis of Algorithm 2, this paper proposes the incremental mining algorithm of a sensitive data recognition function based on global function fitting (ISDR-GFF). Based on the concept of grain granulation, the architecture of the sensitive function model mining by using multicomputing nodes is constructed. A multipopulation grafting operation based on the population similarity is proposed among the computing nodes to improve the population diversity of each computing node and the convergence speed of the GEP.

The parallel sensitive data function model based on grain granulation divides the GEP population to be evolved into several subpopulations. Each subpopulation executes Algorithm 2 in parallel on its respective computing nodes. When the current computing node population similarity is high, the grafting genetic operation among the subpopulations is performed. Therefore, the model can improve the efficiency of the population evolution from a global perspective. Furthermore, it also enriches the diversity of the individuals in the population and controls the premature phenomenon of the sensitive data mining algorithm because of the continuous use of the grafting operation among the subpopulations in the process of evolution. The algorithm model is shown in Fig. 3.

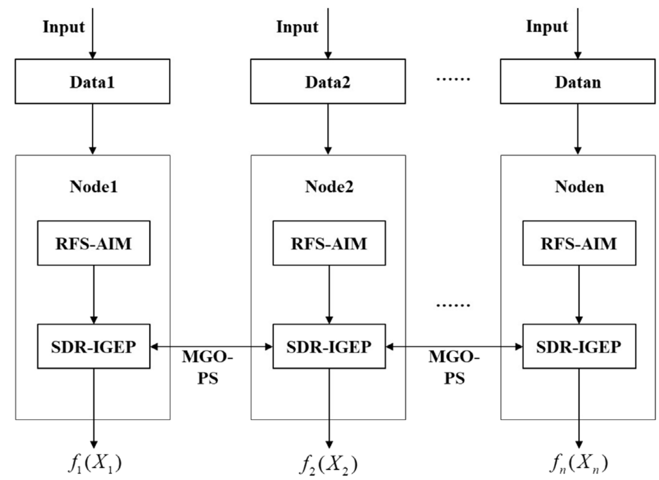

Fig. 3. Parallel sensitive function model based on grain granulation.

Definition 7. Let $Pop_i = (p_{i1}, p_{i2}, \ldots, p_{in})$ and $Pop_j = (p_{j1}, p_{j2}, \ldots, p_{jn})$ represent the current population set on the ith node and jth node, respectively, while $J_c(p_{ik}, p_{jk}) = \frac{|p_{ik} \cap p_{jk}|}{|p_{ik} \cup p_{jk}|}$ represents the similarity between the kth chromosome in the current population set on the ith and jth nodes. Then, $J_p(Pop_i, Pop_j) = \frac{\sum_{k=1}^{n} \frac{|p_{ik} \cap p_{jk}|}{|p_{ik} \cup p_{jk}|}}{n}$ is called the population similarity between the current population Pop_i of the ith node and the current population Pop_j of the jth node.

Where $0 < J_p(Pop_i, Pop_j) < 1$.

On the basis of Definition 7, the population similarity is used to conduct multi-group grafting operations on populations between multiple nodes, increasing the individual diversity in each node population. The multi-population grafting operation based on population similarity (MGO-PS) is described as shown in Procedure 2.

Procedure 2: MGO-PS
Input: Pop_i, Pop_j, $popSize$, p_g;
Output: $newPop_i$, $newPop_j$;
1. for $k \leftarrow 1$ to $popSize$ do
2. $J_c(p_{ik}, p_{jk}) \leftarrow \frac{|p_{ik} \cap p_{jk}|}{|p_{ik} \cup p_{jk}|}$;
3. $J_p(Pop_i, Pop_j) += J_i(p_{ik}, p_{jk})$;
4. end for
5. $J_p(Pop_i, Pop_j) \leftarrow \frac{J_p(Pop_i, Pop_j)}{n}$;
6. if $J_p(Pop_i, Pop_j) > 0.5$ then
7. $newPop_i \leftarrow Graft(p_g, pop_j, popSize)$;
8. $newPop_j \leftarrow Graft(p_g, pop_i, popSize)$;
9. end if
10. Return $newPop_i$, $newPop_j$;

From Fig. 3, it can be seen that the multi-population grafting genetic operation based on population similarity can constantly dynamically adjust the population size on each computing node. Because the offspring of the grafting operation carry the genes of the population on other computing nodes, the diversity of individuals in the population is increased, and the harm of inbreeding is effectively avoided. Meanwhile, the excellent genes in other populations are introduced, which also speeds up the mining efficiency of the sensitive data recognition function model.

5.1. ISDR-GFF

The characteristics of the active distribution network determine the dynamic increase in the various types of data during its operation. To ensure the efficiency of the sensitive data mining on each node shown in Fig. 3, this paper proposes an incremental mining algorithm of the sensitive data recognition function based on global function fitting (ISDR-GFF). The main idea is to start the incremental data judgment program. When the data are added to any node, the algorithm on the current node is started to mine the sensitive data recognition function for the incremental data, and the corresponding incremental sensitive data recognition function model is obtained. Then, the global sensitive data recognition function model is obtained by global fitting with the existing function model, and it replaces the existing function model on the current node. Finally, we use the obtained global sensitive data recognition function to identify the data sensitive type transmitted on the node.

Let $f^i(K)$ be the current sensitive data recognition function model obtained by executing SDR-IGEP on the ith node for dataset $K = (k_1, k_2, \ldots, k_n)$, where $f^i_{new}(K')$ be the new sensitive data recognition function model obtained by performing SDR-IGEP on the ith node for the incremental dataset $K' = (k'_1, k'_2, \ldots, k'_m)$. Here, the dataset $K = (k_1, k_2, \ldots, k_n)$ and $K' = (k'_1, k'_2, \ldots, k'_m)$ have the same dimensions.

Definition 8. For the datasets $K = (k_1, k_2, \ldots, k_n)$ and $K' = (k'_1, k'_2, \ldots, k'_m)$, there are always two constants α and β that are not all zero such that $f^i(K+K') = \alpha f^i(K) + \beta f^i_{new}(K')$ holds. Then, $f^i(K+K')$ is called the global function model after adding dataset $K' = (k'_1, k'_2, \ldots, k'_m)$ onto the ith node.

From Definition 8, when $\alpha \neq 0, \beta = 0$, the global function model $f^i(K+K) = \alpha f^i(K)$, which indicates that the function model obtained from dataset $K = (k_1, k_2, \ldots, k_n)$ can fully fit the incremental dataset $K' = (k'_1, k'_2, \ldots, k'_m)$. When $\alpha = 0, \beta \neq 0$, the global function model $f^i(K+K') = \beta f^i_{new}(K')$, which shows that the function model mined from the incremental dataset $K' = (k'_1, k'_2, \ldots, k'_m)$ can fully fit the dataset $K = (k_1, k_2, \ldots, k_n)$.

To solve $f^i(K+K')$, we only have to solve for the coefficients α and β. According to the goal of polynomial fitting, to make $f^i(K+K')$ fit all of the datasets $K+K'$, we only need to solve for α and β so that Eq. (1) holds.

$$min \sum_{j=1}^{m+n} (y_j - (\alpha f^i(K_j) + \beta f^i_{new}(K'_j)))^2 \quad (1)$$

where $m+n$ is the size of the dataset $K+K'$.

Set $y_j = \chi_j, f^i(K_j) = A_j, f^i_{new}(K'_j) = B_j$ and denote $f(\alpha, \beta) = \sum_{j=1}^{m+n} (\chi_j - \alpha A_j - \beta B_j)^2$. Solving the partial derivatives of $f(\alpha, \beta)$

with respect to α and β, respectively, we have that

$$\begin{cases} \dfrac{\partial f(\alpha,\beta)}{\partial \alpha} = \alpha \sum_{j=1}^{m+n} A_j^2 - \sum_{j=1}^{m+n} \chi_j A_j + \beta \sum_{j=1}^{m+n} A_j B_j \\ \dfrac{\partial f(\alpha,\beta)}{\partial \beta} = \beta \sum_{j=1}^{m+n} B_j^2 - \sum_{j=1}^{m+n} \chi_j B_j + \alpha \sum_{j=1}^{m+n} A_j B_j \end{cases}. \quad (2)$$

To minimize $f(\alpha,\beta)$, Eq. (3) is established

$$\begin{cases} \dfrac{\partial f(\alpha,\beta)}{\partial \alpha} = 0 \\ \dfrac{\partial f(\alpha,\beta)}{\partial \beta} = 0 \end{cases}. \quad (3)$$

By solving Eq. (3), we can obtain the values of α and β that are shown as follows.

$$\begin{cases} \alpha = \dfrac{\sum_{j=1}^{m+n} \chi_j A_j \times \sum_{j=1}^{m+n} B_j^2 - \sum_{j=1}^{m+n} \chi_j B_j \times \sum_{j=1}^{m+n} A_j B_j}{\sum_{j=1}^{m+n} A_j^2 \times \sum_{j=1}^{m+n} B_j^2 - (\sum_{j=1}^{m+n} A_j B_j)^2} \\ \beta = \dfrac{\sum_{j=1}^{m+n} \chi_j A_j \times \sum_{j=1}^{m+n} A_j B_j - \sum_{j=1}^{m+n} \chi_j B_j \times \sum_{j=1}^{m+n} A_j^2}{(\sum_{j=1}^{m+n} A_j B_j)^2 - \sum_{j=1}^{m+n} B_j^2 \times \sum_{j=1}^{m+n} A_j^2} \end{cases}. \quad (4)$$

The global function model can be obtained by substituting the values of α and β in Eq. (4) into $f^i(K+K') = \alpha f^i(K) + \beta f^i_{new}(K')$.

The description of Algorithm 3 is shown as follows.

Algorithm 3: ISDR-GFF

Input: $K, K^{'}, popSize, maxFitness, \mathrm{P_m}, \mathrm{P_t}, \mathrm{P_r}, \beta, \alpha, \mathrm{p_g}, p$;
Output: *GFunction*;
/*For the ith node*/
1. $Pop(i) \leftarrow SDR - IGEP(K, popSize, \max Fitness, P_m, P_t, P_r, \beta, P_g, p, \alpha)$;
2. $f^i(K) \leftarrow SDR - IGEP(K, popSize, \max Fitness, P_m, P_t, P_r, \beta, P_g, p, \alpha)$;

/*For the jth node*/
3. $Pop(j) \leftarrow SDR - IGEP(K, popSize, \max Fitness, P_m, P_t, P_r, \beta, P_g, p, \alpha)$;

/*Exchanging population between the ith node and the jth node*/
4. $newPop(i) \leftarrow MGO - PS(P_g, Pop_j, popSize)$;
5. $newPop(j) \leftarrow MGO - PS(P_g, Pop_i, popSize)$;

/*For the incremental data on the ith node*/
6. $f^i_{new}(K^{'}) \leftarrow SDR - IGEP(K^{'}, popSize, \max Fitness, P_m, P_t, P_r, \beta, P_g, p, \alpha)$;
7. $f^i(K+K^{'}) \leftarrow \alpha f^i(K) + \beta f^i_{new}(K^{'})$;
8. $y_j \leftarrow \chi_j, f^i(K_j) \leftarrow A_j, f^i_{new}(K^{'}_j) \leftarrow B_j$;
9. $f(\alpha,\beta) \leftarrow \sum_{j=1}^{m+n} (\chi_j - \alpha A_j - \beta B_j)^2$;
10. $\frac{\partial f(\alpha,\beta)}{\partial \alpha} \leftarrow 0$;
11. $\frac{\partial f(\alpha,\beta)}{\partial \beta} \leftarrow 0$;
12. $GFunction \leftarrow f(\alpha,\beta)$;
13. Return *GFunction*;

6. Experiment and analysis

6.1. Experimental environment

To better explain the effectiveness and feasibility of the proposed algorithms, the related experiments are performed in a laboratory environment. The hardware in the experiments includes five algorithm servers and one administration server. The hardware and software configurations are shown in Tables 1 and 2, respectively.

Table 1
Hardware configuration in experiments.

No.	CPU	Memory	Hard disk	Function
1	2*E5-2620V2	64G	2*4T,7200K,SATA	Algorithm
2	2*E5-2620V2	64G	2*4T,7200K,SATA	Algorithm
3	2*E5-2620V2	64G	2*4T,7200K,SATA	Algorithm
4	2*E5-2620V2	64G	2*4T,7200K,SATA	Algorithm
5	2*E5-2620V2	64G	2*4T,7200K,SATAt	Algorithm
6	2*E5-2620V2	128G	2*4T,7200K,SATA	Administration

Table 2
Software configuration in experiments.

No.	OS	Programming Language
1	Ubuntu 17.10	Jdk 1.7+Eclipse Luna 4.4
2	Ubuntu 17.10	Jdk 1.7+Eclipse Luna 4.4.2
3	Ubuntu 17.10	Jdk 1.7+Eclipse Luna 4.4.2
4	Ubuntu 17.10	Jdk 1.7+Eclipse Luna 4.4.2
5	Ubuntu 17.10	Jdk 1.7+Eclipse Luna 4.4.2
6	Windows 10 Professional	/

Table 3
Description of the experimental datasets.

Dataset	Name	Instance of dataset	Instance of feature
Training dataset	IEEE 30-Bus	30	13
	HomeA-2014	26280	28
	HomeF-2014	43800	38
Incremental dataset	HomeA-2016	255399	28
	HomeF-2016	510667	38
Testing dataset	IEEE 57-Bus	57	13
	IEEE 118-Bus	118	13
	HomeA-2015	26280	28
	HomeF-2015	43800	38

In Table 1, the algorithm server is used mainly to execute the sensitive data recognition function mining algorithm, and the management server is used mainly to manage the algorithm nodes and schedule the algorithm nodes for the active distribution network data.

6.2. Data source

The experimental data includes mainly the IEEE benchmark datasets and the Home electrical dataset from http://traces.cs.umass.edu/index.php/Smart/Smart. All of the experimental data are divided into a training dataset, incremental dataset and testing dataset, where the training dataset includes the bus data of IEEE 30-Bus benchmark dataset and the Home A-2014 and Home F-2014 power and weather datasets; the incremental dataset includes the power and weather datasets for the Home A and Home F in 2016; the testing datasets include the IEEE 57-bus, 118-bus, Home A-2015 and Home F-2015 power and weather datasets. A description of the experimental dataset is shown in Table 3.

To test the performance of the proposed algorithm, it is first necessary to perform sensitive marking on all of the datasets in Table 3 according to Definition 6, to obtain an experimental dataset that contains sensitive labels as shown in Table 4.

Where the labels include the sensitive and nonsensitive labels in Table 4. A sensitive label is indicated by 1 and a nonsensitive label is indicated by 0, to operate the proposed algorithms in this paper. For example, 1(30) means that there are 30 data instances with sensitive labels in the IEEE 30-Bus dataset. 0(8760) means that there are 8760 data instances with nonsensitive labels in the HomeA-2014 dataset.

Table 4
Description of the experimental datasets with sensitive labels.

Dataset	Name	Instance of dataset	Instance of feature	Label
Training dataset	IEEE 30-Bus	30	13	1(30) 0(0)
	HomeA-2014	26280	28	1(17520) 0(8760)
	HomeF-2014	43800	38	1(35040) 0(8760)
Incremental dataset	HomeA-2016	255399	28	1(246639) 0(8760)
	HomeF-2016	510667	38	1(501907) 0(8760)
Testing dataset	IEEE 57-Bus	57	13	1(57) 0(0)
	IEEE 118-Bus	118	13	1(118) 0(0)
	HomeA-2015	26280	28	1(17520) 0(8760)
	HomeF-2015	43800	38	1(35040) 0(8760)

Table 5
Experimental parameters for the proposed algorithms.

Type of parameter	Value of parameter
Dataset	IEEE 14-Bus,30-Bus,57-Bus and 118-Bus HomeA-2014, 2015 and 2016 HomeF-2014, 2015 and 2016
FunctionSet	$+ - * / SCTEL$
VariableSet	Setting Variables according to datasets
LinkFunction	$+$
Number of gene	3
GeneHead	9
popSize	500
P_m	0.044
P_t of IS	0.3
P_t of RIS	0.3
P_t of GeneIS	0.1
P_r of onepoint	0.3
P_r of twopoint	0.3
P_r of gene	0.1
P_g of grafting	0.3
p of localselection	0.3
α	6
FitnessFunction	$f_i = \sum_{j=1}^{n}(R - \lvert P_{ij} - T_j \rvert)$

After the corresponding processing of the experimental datasets in Table 4, to test the performance of the proposed algorithm, the experimental parameter settings are shown in Table 5.

Where *SCTEL* in the function set represents the trigonometric functions sin, cos, tan and the exponential function exp and the logarithmic function log, as an acronym.

6.3. Evaluation index

Sensitive data recognition based on function mining is essentially classification. To evaluate the performance of the proposed algorithm, the experiment adopts a measurement standard [35] that can evaluate the performance of the classification algorithm to measure the pros and cons of the algorithm. This paper discusses a typical binary classification problem that divides data into sensitive data and nonsensitive data.

If instance I is the sensitive type, then TP is the number of sample data instances that are correctly divided into the sensitive type by the algorithm, and FN is the number of sample data wrongly divided into nonsensitive type by the algorithm. If instance I is the nonsensitive type, then TN is the number of sample data instances of the nonsensitive type that are correctly classified by the algorithm, and FP is the number of sample data instances that are wrongly classified by the algorithm as the sensitive type. We have the following evaluation index.

(1) $p = \frac{TP}{TP+FP}$ is called the precision, which is the proportion of the number of samples correctly classified as sensitive labels to the total number of samples classified as sensitive labels.

(2) $r = \frac{TP}{TP+FN}$ is called the recall, which is the proportion of the number of samples correctly classified as sensitive labels to the total number of samples actually classified as sensitive labels.

(3) $acc = \frac{TP+TN}{TP+FN+FP+TN}$ is called the accuracy, which is the number of correctly classified samples divided by the total number of samples. Generally speaking, the higher the accuracy, the better the classifier.

(4) $spe = \frac{TN}{FP+TN}$ is called the specificity, which represents the proportion of all nonsensitive type instances that are correctly classified, and measures the classifier's ability to recognize nonsensitive type instances.

To evaluate the advantages and disadvantages of the different algorithms, the evaluation index F_1 is proposed based on the precision and recall.

(5) $F_1 = \frac{2pr}{p+r}$ is called the F_1 index. According to the formula, F_1 combines the results of p and r. The higher the value of F_1 is, the more effective the algorithm is.

Last, the parallel performance of the proposed algorithm in this paper is evaluated by the speedup ratio on the premise of keeping the dataset unchanged.

Definition 9. Suppose that the time taken to process dataset D on one node is t_D^1 and the time spent to process dataset D on m nodes is t_D^m. Then, speedup $(m) = \frac{t_D^1}{t_D^m}$ is called the speedup ratio when the number of nodes is m.

6.4. Experimental analysis

Experiment 1. To verify the influence of the feature selection algorithm on sensitive data recognition function mining, this paper first compares the feature selection results between RFS-AIM and the attribution reduction algorithm based on positive region (AR-PR) [36], the attribute reduction algorithm based on a discernible matrix (AR-DM) [37]. Fig. 4 shows a comparison of the number of conditional attributes of the training datasets before and after feature selection based on RFS-AIM, AR-PR and AR-DM. Fig. 5 indicates a comparison of the average running time of the three algorithms while running independently 10 times. To compare the performance of the SDR-IGEP algorithm before and after the feature selection, for all of the training datasets in Table 1, Fig. 6 shows a comparison of the recognition precision, recall and F_1-index.

As seen from Fig. 4, the larger the number of features that the dataset has, and the more significant the effect of the RFS-AIM is. Compared with that before the feature selection, the RFS-AIM algorithm reduced the number of features of the data by approximately 61.54%, 82.14% and 86.84%, respectively, for the IEEE 30-bus, HomeA-2014 and HomeF-2014 datasets. Compared with AR-PR and AR-DM, RFS-AIM reduces the number of features of the HomeA-2014 and HomeF-2014 datasets by 28.57%, 37.5% and 37.5%, 37.5%, respectively. However, for the IEEE 30-bus dataset, the feature selection effect of the RFS-AIM algorithm is not as good as that of the traditional AR-PR and AR-DM. The main reason is that RFS-AIM sorts according to the degree of importance of the conditional features with respect to the classification features, and then, the feature selection is conducted according to artificially set values. However, the AR-PR and AR-DM perform feature reduction based on the positive domain and the discrimination matrix in the rough set. Therefore, for the IEEE 30-bus dataset,

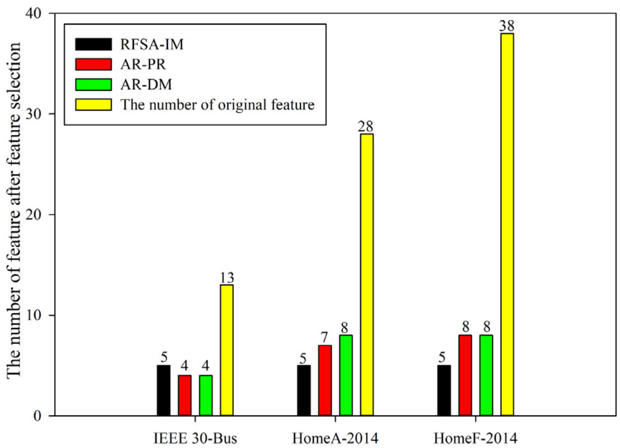

Fig. 4. Comparison of the number of features for all of the training datasets before and after the feature selection.

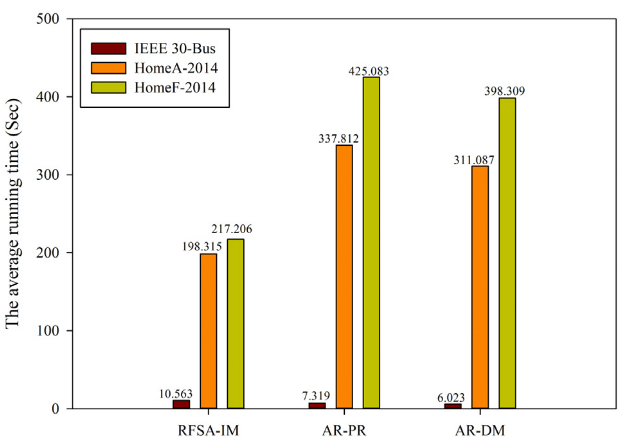

Fig. 5. Comparison of the average running-time for the three algorithms.

due to the small number of conditional features, the number of optimal features selected by AR-PR and AR-DM is better than that of the RFS-AIM algorithm.

Fig. 5 compares the average running time of the three feature selection algorithms. The results show that the smaller the amount of data is, the smaller the number of features, while the average running time of RFS-AIM, AR-PR and AR-DM is almost the same. With an increase in the data volume and feature number, the average running time of RFS-AIM decreases significantly. Compared with AR-PR and AR-DM, the average running time of RFS-AIM is reduced by 48.9% and 45.47%, respectively, for the HomeA-2014 and HomeF-2014 datasets. This result arises because among the three feature selection algorithms, RFS-AIM has the smallest time complexity, approximately $O(|C|\log|C|)$, where $C|$ represents the number of conditional features contained in the dataset.

It can be seen from Fig. 6 that the precision, recall and F_1 index of the training datasets are greatly improved by the feature selection. Compared with before feature selection, the precision of SDR-IGEP is improved by approximately 0%, 19.78% and 28.71%, respectively, for the IEEE 30-bus, HomeA-2014 and HomeF-2014 datasets. The recall of SDR-IGEP increased by 33.33%, 32.41% and 54.43%, respectively. The F_1 index of SDR-IGEP improved by approximately 20.84%, 26.54% and 45.95%, respectively. This result occurs mainly because the feature selection greatly reduces the complexity of the training dataset in such a way that the sensitive data recognition function model based on the SDR-IGEP

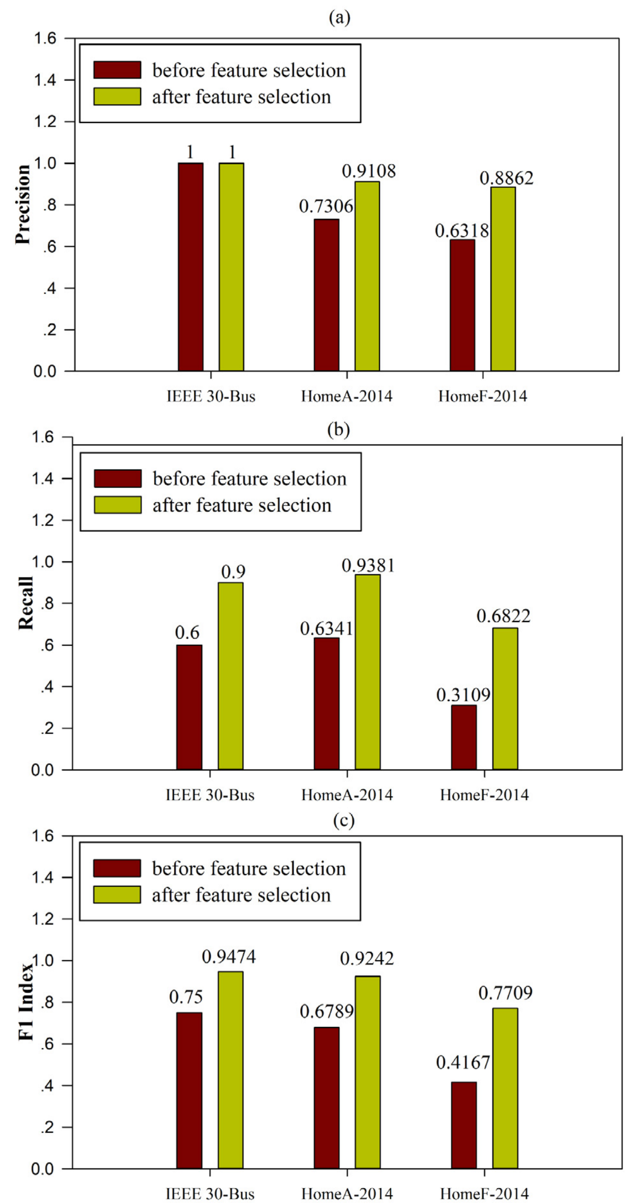

Fig. 6. Precision, recall and F_1 index of SDR-IGEP for all of the training datasets before and after the feature selection.

algorithm is more concise and accurate. We know that sensitive data recognition is essentially binary classification. According to the evaluation standard of the classification algorithm, the higher the value of the F_1 index is, the better the classification effect of the algorithm. Therefore, the experimental results in Fig. 6(c) indicate that feature selection can indeed improve the accuracy of SDR-IGEP in recognizing sensitive data.

Experiment 2. Sensitive data recognition is essentially binary classification. This experiment compares SDR-IGEP with traditional GEP (T-GEP) [11], GP [38], SVM [39]and KNN [40] in the precision, recall, F_1 index, accuracy, sensitive and specificity of sensitive data recognition after the feature selection. The parameter of T-GEP is similar to SDR-IGEP. And the parameter of GP, SVM and KNN is shown in Tables 6–8. For all of the training datasets in Table 1, Fig. 7 shows a comparison of SDR-IGEP, T-GEP, GP, SVM and KNN in the recognition precision, recall, F_1 index, accuracy and specificity. For all of the testing datasets in Table 1, Fig. 8

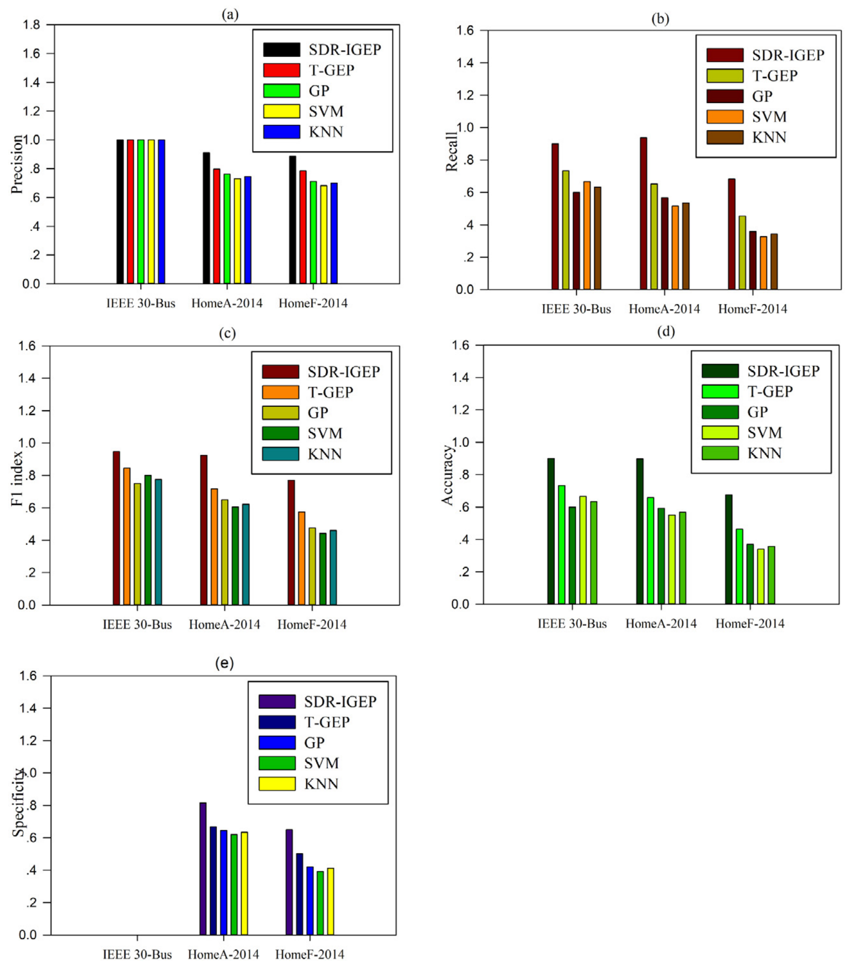

Fig. 7. Precision, recall, F_1 index, accuracy and specificity based on SDR-IGEP, T-GEP, GP, SVM and KNN for all of the training datasets.

Table 6
The parameters of GP.

Type of parameter	Value of parameter
Dataset	IEEE 14-Bus,30-Bus,57-Bus and 118-Bus HomeA-2014, 2015 and 2016 HomeF-2014, 2015 and 2016
popSize	500
Generation	40
Tournament Selection	7
P_c	0.5
P_m	0.1

Table 7
The parameters of SVM.

Type of parameter	Value of parameter
Dataset	IEEE 14-Bus,30-Bus,57-Bus and 118-Bus HomeA-2014, 2015 and 2016 HomeF-2014, 2015 and 2016
Kernel-type	RBF
Degree	3.0
Gamma	0.5
b	0.5
C	20

shows the prediction accuracy of the sensitive types based on SDR-IGEP, T-GEP, GP, SVM and KNN.

It can be seen from Fig. 7 that, for the IEEE 30-bus, HomeA-2014 and HomeF-2014 datasets, SDR-IGEP and T-GEP are superior to GP, SVM and KNN in their precision, recall rate , F_1 index and accuracy. However, for the IEEE 30-Bus dataset, the specificity of all algorithms is equal to 0 because the number of non-sensitive labels contains 0. And for HomeA-2014 and HomeF-2014 datasets, the SDR-IGEP algorithm has obvious advantages. And compared with the T-GEP algorithm, for the IEEE 30-bus, HomeA-2014 and HomeF-2014 datasets, the precision of SDR-IGEP is improved by 0%, 12.47% and 11.47%, respectively; the recall of SDR-IGEP is increased by 18.56%, 30.42% and 33.47%, respectively; the F_1 index of SDR-IGEP is increased by 10.71%, 22.34% and 25.4%, respectively; the accuracy of SDR-IGEP is increased by 18.56%, 26.64% and 31.51%, respectively; and the specificity of SDR-IGEP is increased by 0%, 18.14% and 22.92%, respectively. Based on the evaluation indexes of the classification algorithm, compared with other algorithms, the SDR-IGEP algorithm has higher precision and accuracy. At the same time, according to the recall and specificity, experimental results also show that whether it is sensitive or non-sensitive data, The SDR-IGEP can accurately identify. The good recognition performance of

Table 8
The parameters of KNN.

Type of parameter	Value of parameter
Dataset	IEEE 14-Bus,30-Bus,57-Bus and 118-Bus HomeA-2014, 2015 and 2016 HomeF-2014, 2015 and 2016
K	2
Calculation of distance	Euclidean distance
Decision rules	Majority-voting

Table 9
Description of the incremental dataset.

Type	Name	Instance of dataset	The number of features	Label
Training dataset	HomeA-2016-training	200000	28	1(19300) 0(7000)
	HomeF-2016-training	300000	38	1(294600) 0(5400)
Testing dataset	HomeA-2016-testing	55399	28	1(53639) 0(1760)
	HomeF-2016-testing	210667	38	1(207307) 0(3360)

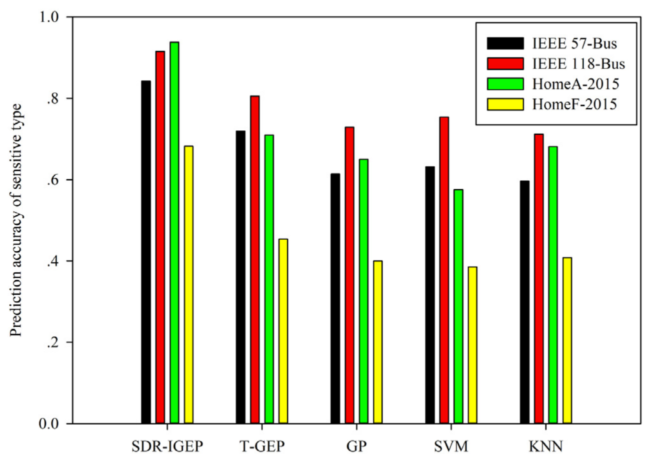

Fig. 8. Prediction accuracy of sensitive type based on SDR-IGEP, T-GEP, GP, SVM and KNN for all of the testing datasets.

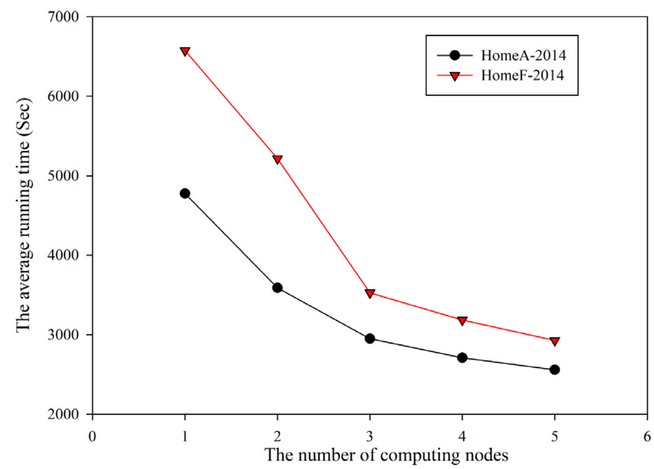

Fig. 9. Comparison of the average running time based on ISDR-GFF under different numbers of nodes.

SDR-IGEP occurs mainly because the algorithm discards the chromosomes that have higher similarity in the population during the operation, which increases the diversity of the chromosomes in the population and avoids the local convergence of the algorithm. Furthermore, we also find that the recognition precision of the four algorithms is always 1 for the IEEE 30-Bus dataset. This finding occurs because the IEEE 30-Bus dataset contains only sensitive data types and does not contain nonsensitive data types, which leads to an imbalance in the data types, in such a way that the value of *FP* is always 0 when we calculate the precision.

For the IEEE 57-Bus, IEEE 118-Bus, HomeA-2015 and HomeF-2015 testing datasets, Fig. 8 shows that SDR-IGEP has reached 84.21%, 91.53%, 93.81% and 68.21% in the recognition accuracy of the sensitive types, respectively. It also shows that SDR-IGEP and T-GEP have better prediction accuracy than GP, SVM and KNN in identifying the sensitive types. Specifically, compared with T-GEP, SDR-IGEP improves the accuracy of the sensitive type recognition by 14.58%, 12.04%, 24.35% and 33.46%, respectively, for the IEEE 57-Bus, IEEE 118-Bus, HomeA-2015 and HomeF-2015 testing datasets. This finding also indicates that SDR-IGEP can effectively identify sensitive data. Through the identification and modeling of the IEEE 30-Bus, HomeA-2014 and HomeF-2014 training datasets, other sensitive datasets that contain corresponding characteristics can be well identified, which provides methodological support for sensitive recognition of other numerical business system data in active distribution networks.

Experiment 3. According to the number of computing nodes, the HomeA-2014 and HomeF-2014 training datasets in Table 1 are equally divided into several sub-datasets. Fig. 9 compares the average running time of ISDR-GFF under different computing nodes. At the same time, the first 200,000 pieces of the incremental dataset HomeA-2016 are composed of the training dataset HomeA-2016-training; the first 300,000 pieces of the HomeF-2016 data constitute the training dataset HomeF-2016-training, and the remainder are composed of the testing datasets HomeA-2016-testing and HomeF-2016-testing, as shown in Table 9. Then, the HomeA-2016-training dataset is deployed to one of the computing nodes. Fig. 10 shows a comparison of the recognition precision, recall rate and F_1 index between ISDR-GFF and SDR-IGEP before and after adding the HomeA-2016-training and HomeF-2016-training datasets. For all of the testing datasets in Table 9, the accuracy prediction of the sensitive types based on ISDR-GFF is given in Fig. 11. Moreover, to test the parallel performance of the ISDR-GFF, Fig. 12 shows the speedup ratio of the ISDR-GFF when the number of computing nodes increases.

Fig. 9 shows that with the increasing number of computing nodes, for the incremental datasets, the average running time of ISDR-GFF decreases continuously. This finding occurs mainly because in the laboratory environment, the information interaction time between the various nodes is negligible in the calculation time-consumption of the ISDR-GFF. In addition, the hardware performance of all of the computing nodes in the experiment is exactly the same, and thus, the calculation delay of each computing node is also negligible. Meanwhile, it can be seen from Fig. 9 that when the number of computing nodes increases from 1 to 3, the average running time of the ISDR-GFF decreases the fastest; and when the number of computing nodes increases from 3 to 5, the average running time of the ISDR-GFF slows down. This finding occurs because when the number of computing nodes increases to a certain number, the time gap of the datasets processed on each computing node becomes smaller and smaller,

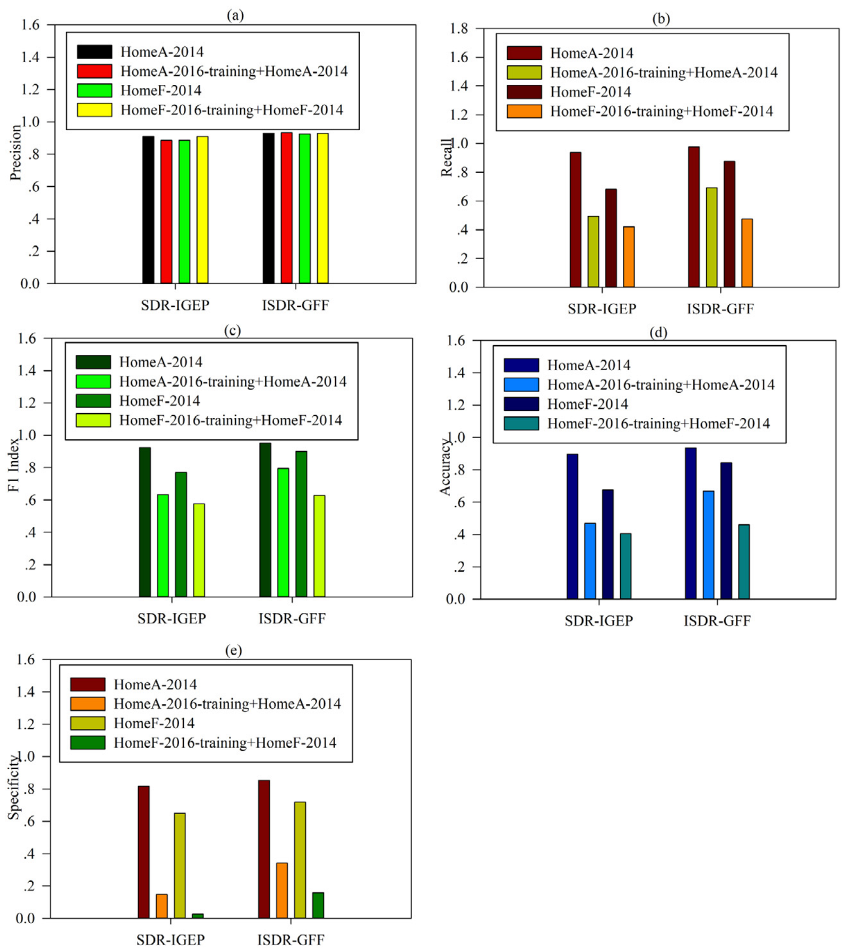

Fig. 10. Comparison of the precision, recall and F_1 index, accuracy and specificity between ISDR-GFF and SDR-IGEP before and after adding the incremental training datasets.

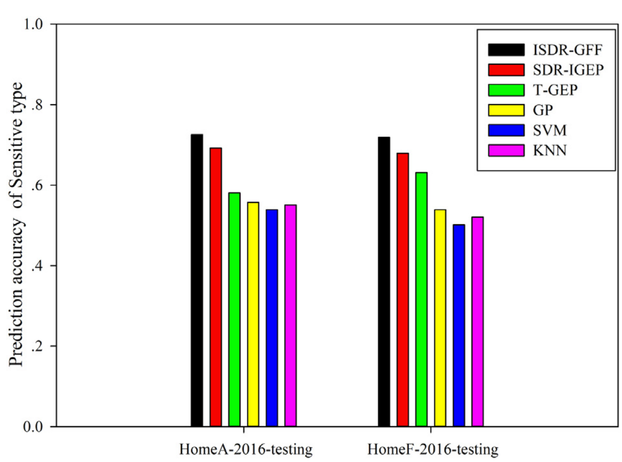

Fig. 11. Prediction accuracy of the sensitive type between ISDR-GFF, SDR-IGEP, T-GEP, GP, SVM and KNN for all of the incremental testing datasets.

which slows down the average running time's downward trend of the ISDR-GFF.

As seen from Fig. 10, the precision, recall, F_1 index, accuracy and specificity of ISDR-GFF are superior to the SDR-IGEP, regardless of whether the incremental training dataset is added. Furthermore, when new datasets are added, the precision, recall, F_1 index, accuracy and specificity of the SDR-IGEP and ISDR-GFF all significantly decrease. This finding occurs because when the incremental training dataset is added, for the SDR-IGEP, the volume of training datasets is greatly increased, this will affect SDR-IGEP's classification model for identifying sensitive data. However, for ISDR-GFF, it is difficult to solve to make the global sensitive data recognition model reach the global optimal in the process of generating the global sensitive data recognition model. For the incremental datasets, the specificity of SDR-IGEP and ISDR-GFF is not significant. The result demonstrate that SDR-IGEP and ISDR-GFF cannot classify the non-sensitive data of the incremental datasets well. This may be because the number of data instances belonging to non-sensitive label in the incremental datasets is much less than the data instances belonging to sensitive label, thus causing the problem of unbalanced classification. When the HomeA-2016-training and HomeF-2016-training datasets are added in the experiment, compared with SDR-IGEP, the precision, recall, F_1 index, accuracy and specificity of ISDR-GFF are improved by 5.12%, 28.69%, 20.25%, 29.64%, 56.89%, 2.1%, 11.27%, 8.36%, 11.74% and 83.65%, respectively. This finding indicates that the ISDR-GFF has good adaptability and high classification recognition ability for incremental datasets.

Fig. 11 shows that for all incremental testing datasets, the sensitive type prediction accuracy of ISDR-GFF is better than SDR-IGEP, T-GEP, GP, SVM and KNN. Specifically, compared with SDR-IGEP, T-GEP, GP, SVM and KNN, for the HomeA-2016-testing dataset, the sensitive type prediction accuracy of ISDR-GFF is

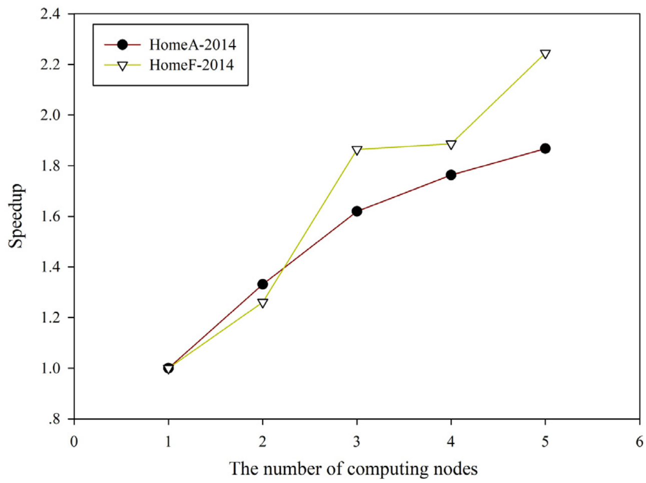

Fig. 12. Speedup of ISDR-GFF with different numbers of computing nodes.

improved by 4.54%, 19.92%, 23.16%, 25.64% and 23.96%, respectively; For the HomeF-2016-testing dataset, the sensitive type prediction accuracy of ISDR-GFF is improved by 5.43%, 12.11%, 25.01%, 30.17% and 27.53%, respectively. This finding also demonstrates that the ISDR-GFF has better recognition and prediction capabilities on sensitive types for the incremental dataset.

Speedup is an important index to measure the performance of parallel algorithms. Fig. 12 demonstrates that with the increase in computing nodes, the speedup of ISDR-GFF gradually increases for the HomeA-2014 and HomeF-2014 datasets. Especially, when the number of computing nodes increases from 1 to 3, the speedup of the ISDR-GFF increases significantly; and when the number of nodes increases from 3 to 5, the speedup of the ISDR-GFF increases slowly. We know that the ideal speedup should be linear. However, from Fig. 12, we can see that for the HomeA-2014 dataset, the speedup is basically linearly accelerated, and for the HomeF-2014 dataset, the speedup is not a linear rise. This finding occurs mainly because as the number of computing nodes increases, datasets are evenly distributed in the increases, in such a way that the time for exchanging information between different computing nodes increases, which makes a linear acceleration ratio difficult to achieve.

Through the above three experiments, the feature selection algorithm greatly reduces the complexity of the numerically sensitive data in the active distribution network. On this basis, the sensitive data recognition model based on improved gene expression programming can well measure whether the various numerical business data in an active distribution network can safely achieve the designated destination network or host through the network. Thus, a series of operations on active distribution network control have been safely completed, and finally, the results ensure the safe and stable operation of various business systems in an active distribution network.

7. Conclusions

The safe and efficient transmission of data is critical to the safe and stable operation of active distribution network business systems. The existing data transmission security protection in an active distribution network focuses on unstructured and structured data such as text and database, and is mainly based on centralized text classification algorithms and traditional security defense methods such as access control and encryption. Therefore, to solve the intelligent recognition of dynamically increasing numerical sensitive datasets in active distribution networks, first, a rough feature selection algorithm based on an average importance measurement (RFS-AIM) is proposed to quantitatively analyze the average importance of each reduced feature to the final decision attribute. Second, based on the RFS-AIM, a sensitive data recognition function mining algorithm based on feature selection and improved gene expression programming (SDR-IGEP) is proposed. In the SDR-IGEP algorithm, the concept of chromosome similarity is used to improve the genetic operation in the traditional gene expression programming algorithm to avoid the GEP population falling into local optimum. Then, an incremental mining algorithm of a sensitive data recognition function based on global function fitting (ISDR-GFF) is proposed to solve recognition of the incremental sensitive data. In addition, the framework of multi-computing node parallel function model mining is constructed based on grain granulation. Furthermore, a multi-population grafting genetic operation based on population similarity is proposed to improve the population diversity of the computing nodes and the convergence speed of the GEP population. Last, for the IEEE-30, 57, 118 bus systems benchmark datasets and real power load data, the simulation results show that the proposed algorithm has certain advantages over the traditional algorithm in terms of the sensitive data recognition precision, recall, F_1 index, average running time and speedup.

The experimental results show that the algorithm is sensitive to the number of sensitive types in the simulation data. More specifically, for datasets with uneven sensitive types and less sample data (e.g., IEEE 30-Bus, 57-Bus and 118-Bus), the algorithm will lead to overfitting of the sample data due to a lack of representative samples. We know that this type of dataset is common in a real active distribution network environment. Thus, it is necessary to design the corresponding algorithms to avoid model overfitting in the future to truly reflect the model contained in the sample data. At the same time, it should be noted that the algorithms proposed in this paper cannot be applied to numerical datasets in all fields. And when reducing the features of the dataset, in addition to using the rough set proposed in this work, you can also use principal component analysis, clustering, etc.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Song Deng: Conceptualization, Methodology, Software, Writing - original draft, Formal analysis, Funding acquisition. **Xiangpeng Xie:** Data curation, Writing - review & editing. **Changan Yuan:** Resources, Investigation. **Lechan Yang:** Investigation. **Xindong Wu:** Project administration, Supervision.

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Appendix. List of abbreviations

RFS-AIM	A Rough Feature Selection algorithm based on the Average Importance Measurement
SDR-IGEP	A Sensitive Data Recognition Function Mining algorithm based on RFS-AIM and Improved Gene Expression Programming
ISDR-GFF	A New Incremental Mining algorithm for a Sensitive Data Recognition Function based on Global Function Fitting
SCADA	Supervisory Control And Data Acquisition
DMS	Database Management System
AMI	Advanced Metering Infrastructure
PUP-CS	Population Update based on Chromosome Similarity
MGO-PS	Multipopulation Grafting Operation based on Population Similarity
AR-PR	The Attribution Reduction algorithm based on Positive Region
AR-DM	The attribute reduction algorithm based on a discernible matrix
GEP	Gene Expression Programming
T-GEP	Traditional GEP
GP	Genetic Programming
SVM	Support Vector Machine

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